

Assignment-1

Cloud Fundamentals – AWS

We mainly focused on three important AWS services:

1. **Computing (EC2 instances, lambda)**
 2. **Storage (S3 buckets, EBS, EFS)**
 3. **Networking (VPC, Route S3)**
 4. **Identity and Access Management (IAM)**
 5. **Security (WAF, KMS)**
- **Region** – A geographical area where AWS has its data centers. Eg: Hyderabad.
 - **Availability Zone (AZ)** – A separate data center location within a Region. Eg: Northern , Southern areas of Hyderabad and so on.
1 Region can have 2+AZs.

COMPUTING:

What is an **EC2 instances**?

EC2 instances are virtual servers in AWS that we can create and use to run applications and other workloads in the cloud.

AMI (Amazon Machine Image) -> a template used to create EC2 instances.

EC2 Instance -> Virtual server created using that template

Auto-Scaling => increase or decrease the number of EC2 instances based on demand

STORAGE:

1. **Object Storage – S3**
2. **Block Storage – EBS**
3. **File Systems – EFS**

Object Storage (S3 Buckets): -> Stores objects/files

S3 buckets are containers in AWS used to store and organize files such as images, videos, documents, and backups in the cloud.

Block Storage (EBS): -> Provides disk storage for one EC2

EBS is a storage service in AWS that provides persistent storage for EC2 instances, similar to a hard disk attached to a computer.

File Systems (EFS): -> Provides shared file storage for multiple EC2s

EFS is an AWS file storage service that allows multiple EC2 instances to share and access the same files at the same time.

S3 Storage Classes:

1. **S3 Standard** – suitable for frequently accessed data.
2. **S3 Standard-IA** – suitable for data that is accessed less frequently.
3. **S3 Glacier** – suitable for long-term archival data.

S3 Lifecycle Policies -> helps in reducing storage costs and managing data efficiently

NETWORKING:

- VPC (Virtual Private Cloud) - Private AWS network
- Subnets - Smaller sections of a VPC
- Security Groups - Controls access
- Internet connectivity - Connects to the internet

IDENTITY AND SECURITY:

IAM – Identity and Access Management => used to control who can access AWS resources and what they are allowed to do. Basically acts like an identity of the user.

